Lakshya Jindal

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Education

International Institute of Information Technology, Hyderabad Aug 2023 – Present
B.Tech + MS(By Research) in Electronics and Communication Engineering *Hyderabad, India*

Hiranandani Foundation School Thane June 2021 – June 2023
ISC, PCM and Computer Applications - 94.25% *Mumbai, India*

Experience

Undergraduate Researcher | Robotics Research Centre, IIIT Hyderabad April 2025 – Present
Research Assistant *On-site*

- Engineered high-performance numerical optimization algorithms in C++ and Python, focusing on computational latency reduction, memory footprint minimization, and numerical stability.
- Constructed modular testing suites and evaluation pipelines in Linux/Git to systematically benchmark convergence behavior, profile execution performance, and verify code regression.
- Collaborated with researchers to transition experimental prototypes into production-quality C++ software through code reviews, rigorous unit testing, and structured debugging.

Publications

Towards Kinematic Actionable Infeasibility Detection in Motion Planning | arXiv:2609.32806 IEEE RA-L

- Co-authored research introducing a geometry-driven framework for certifying infeasibility in high-dimensional motion planning, leveraging GPU-accelerated simplicial reconstruction and algorithmic optimization.

Projects

Custom Unix-like Shell | *C, POSIX APIs, Systems Programming, Linux, GDB* | GitHub

- Architected a modular POSIX-compliant shell in C featuring process lifecycle management (`fork`, `execvp`), multi-stage piping, I/O redirection, and asynchronous signal handling.
- Enforced strict pointer lifecycle tracking and isolated dispatchers from parsing logic to prevent memory leaks and dangling pointers, validating memory safety using Valgrind.

RISC-V Processor Design & Verification | *SystemVerilog, RTL, Computer Architecture* | GitHub

- Designed a 5-stage pipelined RISC-V processor incorporating hazard detection, data forwarding, and branch resolution logic to minimize pipeline stalls and maximize instruction throughput.
- Developed verification testbenches to validate functional execution correctness across complex control hazards, structural hazards, and arithmetic instruction sequences.

NaviSight – Low-Latency Edge Inference Pipeline | *C++, Python, Edge AI, Optimization* | Website

- Constructed an optimized offline vision inference pipeline in C++ and Python, streamlining runtime execution and memory allocation for resource-constrained edge hardware.
- Profiled compute latency bottlenecks and integrated dynamic audio feedback routines to achieve responsive real-time execution speeds on low-power architectures.

Crazy Taxi – AI Multimodal Claims Assessment | *PyTorch, YOLO, Grad-CAM, NLP* | GitHub

- Developed an end-to-end multimodal pipeline in Python to evaluate claim severity, processing visual and textual data streams via deep convolutional architectures to quantify damage.

Technical Skills

Languages: C, C++ (C++14/17/20), Python, SystemVerilog, Bash/Shell Scripting, SQL
Systems & Compiler Concepts: Computer Architecture, CPU Pipelining, Hazard Resolution, Memory Hierarchy, Cache Locality, POSIX System Calls, Performance Optimization
Developer Tools & Infrastructure: Linux/Unix, GCC, Clang/LLVM, GDB, Valgrind, Git, Make/CMake, VS Code
Core CS & Math: Data Structures & Algorithms, Object-Oriented Programming (OOP), Linear Algebra, Probability & Statistics

Achievements/Extracurriculars

- Solved 200+ algorithms and data structures problems across competitive platforms (Codeforces peak rating: **1269**, Handle: **Laksh_J**).
- Secured **3rd Place** out of 100+ teams in **Megathon 2025** for developing an automated multimodal damage assessment tool.
- Secured **AIR 4391** in JEE Mains 2023 & **AIR 10931** in JEE Advanced 2023; Cleared **KVPY 2022** & **Pre-RMO 2019**.

Relevant Coursework

Intro to Processor Architecture | Operating Systems and Networks | Data Structures and Algorithms | Object - Oriented Programming | Statistical Methods in AI | Computer Vision | Probability & Random Processes